FRANCK JUNIOR ABOYA MESSOU

Doctoral Researcher | AI / LLM for Smart Grids & IoT | Full-Stack & AI Engineering

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PROFESSIONAL SUMMARY

PhD researcher in Applied Informatics at Hosei University (YuLab, IIST Koganei), specialising in **AI, small / large language models (sLM / LLM), and IoT** methods for smart-grid stability, multi-impairment analysis, and SCADA anomaly detection. First-author of **2 Q1 journal papers** (*Internet of Things*, *Expert Systems with Applications*) and **5 IEEE flagship conference papers** (VTC, GLOBECOM, ICC, ICCE). Build and operate **dedicated multi-GPU servers** end-to-end: CUDA toolchain, distributed training, and inference serving. Beyond research, ship production AI and full-stack systems: cross-platform mobile apps, REST APIs, and IoT-integrated pipelines for industry clients.

EXPERIENCE

Full-Stack Developer & AI (Part-time)

D'isum Inc.

Oct 2025 – Jun 2026 | Tokyo, Japan

- **Sensor Data Pipeline Architect & Analyst**: built pipelines from raw IoT sensor data to analytics-ready datasets.
- **Sensor Data Specialist**: app development, prototyping, integration: full scope of development, architecture, and team collaboration on core software integration.
- **Precision Sensor Data Engineer, Analytics, ML Prototypes & System Fusion**: incorporating ML and predictive elements, prototyping, and integrating new systems with existing ones.
- **Sensor apps (iOS & Android)**: built with Java and Swift.

Doctoral Researcher

YuLab, Hosei University (Advisor: Prof. Keping Yu, Stanford / Elsevier World's Top 2% Scientists 2025)

Oct 2025 – Present | Tokyo, Japan

- First-authored **2 Q1 journals** (*Internet of Things* IF 7.6, *Expert Systems with Applications* IF 7.5) and **5 IEEE flagship papers** (VTC, GLOBECOM, ICC, ICCE).
- Design dual-domain **graph-neural-network architectures** for power-system stability at scale (up to 2,869 buses, PEGASE benchmark).
- Develop the **spectral-to-language reprogramming** pipeline (Spec2Llama, Spec2LLM, LLM4Imp) for both LLM and small-language-model backbones.
- Provision and operate a **dedicated multi-GPU server** (CUDA, drivers, distributed training, inference serving).

Master's Researcher

YuLab, Hosei University

2023 – Sep 2025 | Tokyo, Japan

EDUCATION

PhD, Applied Informatics

Hosei University

2023 – Apr 2027 (early) Tokyo, Japan

Thesis direction: AI / sLM / LLM methods for smart-grid stability, multi-impairment analysis, and SCADA anomaly detection on IoT-instrumented power systems.

MSc, Applied Informatics

Hosei University

2023 – Sep 2025 | Tokyo, Japan

Master's research on transformer-based forecasting for IoT energy systems (TAPformer, TSFormer, CNN-BiLSTM hybrid).

TECHNICAL SKILLS

AI / ML & LLMs

PyTorch | Hugging Face | vLLM | scikit-learn | OpenAI / GPT | Anthropic / Claude | Llama | Qwen | Phi | SmolLM

Methods

Time-series forecasting | Spectral reprogramming | Graph neural networks | Federated learning | Prompt engineering

Languages

Python | Java | Dart | SQL | Bash | JavaScript | LaTeX

Backend, web & mobile

Flask | FastAPI | Node.js / Express | GraphQL | REST | Flutter / Dart | React | HTML5 / CSS3

Data & DevOps

MongoDB | PostgreSQL | MySQL | Docker | GitHub Actions | GCP | Linux & Multi-GPU CUD

- Foundational work on **attention-based transformers** for IoT load forecasting: TAPformer, TSFormer, hybrid CNN-BiLSTM that won the VTS Student Paper Award.
- Set up the lab GPU-server training pipeline that the doctoral work now extends.

Intern, GenAI for Business · TICAD9 Booth

Japan AI Consulting (J-AIC)

Jul 2024 · Aug 2025 Osaka · Yokohama

- Internship (Jul 2024): GenAI prototyping for the consulting practice; LLM-powered solutions for real client problems.
- TICAD9 booth (Aug 2025): presented applied ML and GenAI prototypes at JICA's TICAD9 booth (Yokohama).

Teaching & Research Assistant

IIST Department, Hosei University

2024 – Present Tokyo, Japan

Supporting professors in English-language classes: Project-Based Learning (PBL) and graduation research for 4th-year undergraduates.

PUBLICATIONS

- **MFJ Aboya** et al., "TAPformer: Long-Term Electricity Load Forecasting in IoT Systems," *Internet of Things*, Elsevier, 2026. (IF 7.6, Q1)
- **MFJ Aboya** et al., "Spec2Llama: Spectral-Aware Forecasting for Versatile Time Series," *Expert Systems with Applications*, 2026. (IF 7.5, Q1)
- **MFJ Aboya** et al., "Hybrid Attention-Based CNN-BiLSTM for Short-Term Load Forecasting in IoT," *IEEE VTC2024-Fall*, Washington D.C. **Student Paper Award**
- **MFJ Aboya** et al., "Spec2LLM: Spectral-to-Language Reprogramming for Power Transformer Temperature Forecasting," *IEEE ICCE 2026*, Dubai.
- **MFJ Aboya** et al., "LLM4Imp: Frozen LLMs with Spectral Prompts for Time-Series Imputation," *IEEE ICC 2026*, Glasgow.

PROJECTS

- **Smart-grid co-optimization**: dual-domain GNN for power-system stability at PEGASE-2,869-bus scale (PhD).
- **Spec2Llama / Spec2LLM / LLM4Imp**: spectral-to-language reprogramming for LLM and sLM backbones.
- **EKKOU platform**: full-stack AI platform with GPU model serving.

Public code: github.com/mesabo

AWARDS & SERVICE

IEEE VTS Tokyo / Japan Chapter
Student Paper Award & Encouragement Award, VTC2024-Fall (Oct 2024)

JICA Development Studies Program
Graduate, 2025

Review Committee, IEEE ICECET 2026
Rome, IEEE Italy Section

Reviewer, Theoretical Computer Science
Elsevier (ISSN 0304-3975)

LANGUAGES

French	●●●●●
English	●●●●●
Japanese	●●○○○